

EDUCATION

Northeastern University, Boston MA

Ph.D. student in Computer Science

Advisor: Prof. [Arjun Guha](#)

May 2025 - Present

Wellesley College, Wellesley MA

B.A. in Computer Science & Architecture | GPA: 3.98/4.0

Aug 2021 - Dec 2024

Massachusetts Institute of Technology, Cambridge MA

Cross-registered Student | GPA: 5.0/5.0

Jan 2022 - Dec 2024

RESEARCH INTEREST

My research focuses on large language models for code generation and agentic programming. I work on training and evaluating these models across a range of software tasks and programming languages, with the goal of deploying them safely. I draw on insights from agent security, mechanistic interpretability, and human-agent interaction.

PUBLICATION

- [7] [ALIBI: Reasoning Failure Attacks on LLM-Based Vulnerability Detectors via Adversarial Code Comments](#)
[Zixuan Wu](#), Cristina Nita-Rotaru
Preprint. In submission to USENIX 2027
- [6] [The Best Programming Language for Tokenmaxxing: An Investigation of Coding Agent Behavior Across Programming Languages](#)
[Zixuan Wu](#), Carolyn Jane Anderson, Arjun Guha
LMPL 2026. International Workshop on Language Models and Programming Languages at SPLASH 2026. Under review at ACL Rolling Review
- [5] [AgentPack: A Dataset of Code Changes, Co-authored by Agents and Humans](#)
Yangtian Zi, [Zixuan Wu](#), Aleksander Boruch-Gruszecki, Jonathan Bell, Arjun Guha
EMNLP Findings 2026. Findings of the 2026 Conference on Empirical Methods in Natural Language Processing
- [4] [Agnostics: Learning to Synthesize Code in Any Programming Language with a Universal Reinforcement Learning Environment](#)
Aleksander Boruch-Gruszecki, Yangtian Zi, [Zixuan Wu](#), Tejas Oberoi, Carolyn Jane Anderson, Joydeep Biswas, Arjun Guha
ICLR 2026. The Fourteenth International Conference on Learning Representations
- [3] [ReasoningWeekly: A General Knowledge and Verbal Reasoning Challenge for Large Language Models](#)
[Zixuan Wu](#), Francesca Lucchetti, Aleksander Boruch-Gruszecki, Jingmiao Zhao, Carolyn Jane Anderson, Joydeep Biswas, Federico Cassano, Arjun Guha
AAACL-IJCNLP 2025. The 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics
- [2] [Substance Beats Style: Why Beginning Students Fail to Code with LLMs](#)
Francesca Lucchetti, [Zixuan Wu](#), Arjun Guha, Molly Q Feldman, Carolyn Jane Anderson
NAACL ORAL 2025. The 2025 Annual Conference of the Nations of the Americas Chapter of the ACL.

[1] [GlyphPattern: An Abstract Pattern Recognition for Vision-Language Models](#)
[Zixuan Wu](#), Yoolim Kim, Carolyn Jane Anderson
ACL Findings 2025. 63rd Annual Meeting of the Association for Computational Linguistics.

EXPERIENCE

Northeastern Programming Languages Lab, Boston MA

Pre-Doctoral Research Assistant	Feb 2025 - May 2025
Student Researcher	May 2024 - Jan 2025

- Reasoning Models Solving Verbal Puzzles

- Develop and maintain a live benchmark of frontier reasoning models on NPR Sunday Puzzles.
- Identify failure modes ARC-AGI misses, such as giving up, inconsistency, and over-reasoning.

- National Deep Inference Fabric

- Engineer code-free [NNsight](#) interfaces for non-experts to use logit lens and tuned lens.

- Code LLM Interpretability

- Study how LLMs internally represent pointers and aliases in code.
- Construct pointer benchmarks by A-normalizing NetworkX graph code of varying complexity.
- Use path-swapping patching to test if models store intermediate results in register tokens.

- Beginner Student – Code LLM Prompting Study

- Conduct counterfactual causal intervention experiments via tagging and substitution.
- Design a tool for CS professors to see students' knowledge gaps via prompt-editing graphs.

FELLOWSHIP & LEADERSHIP EXPERIENCE

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|---|-------------|
| • NCAA Division III Academic All-Conference Wellesley Varsity Swimming Team | 2021 - 2024 |
| • Teaching Assistant Wellesley Mathematics, Computer Science Department | 2022 - 2024 |
| • Volunteering Program Manager Wellesley Robogals Global | 2022 - 2023 |
| • Slade Summer Fellowship awardee Tanner Conference Speaker Wellesley Art Department | 2022 - 2023 |

TECHNICAL SKILLS

Programming Languages: Python, Java, JS/TS, C/C++, Golang, R, MATLAB. (in level of proficiency)

Development Tools: PyTorch, CUDA, NCCL, HTML/CSS, React, React Native, Node.js, Flask, MySQL, MongoDB, Springboot, OpenCV, Git, Docker, AWS EC2, Figma.